SHI International Corp.

Valtis Oncology Advisory Board – Productionization (Phase 1)

Statement of Work

For AstraZeneca Pharmaceuticals LP

SHI International Corp.
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1 Executive Summary

AstraZeneca Pharmaceuticals LP (“Customer”) has engaged SHI International Corp. (“SHI”) to productionize the Valtis Oncology Advisory Board platform – delivering a secure AWS deployment, single sign-on, a governed frontdoor, and grounded program-deck/competitive-brief generation with export, as Phase 1 of a three-phase roadmap (“Services”).

The division of responsibility is explicit: the Valtis scientific core – the knowledge layer, workflows, output generators, prompts, rubrics, reasoning framework and the simulated advisory board (vSAB) module – comprises existing Customer assets. SHI deploys, integrates, verifies and operates that platform; SHI does not rebuild the scientific core. Accordingly the application layer covered by F10–F17 is delivered as integrate-and-verify work against Customer-provided components, and the vSAB is deployed and demonstrated in full rather than reconstructed.

Three layers produce every revealed artifact: generation from Customer's approved models behind the AI-Gateway; verification by the engine's deterministic fail-closed gates; and evidence from the provenance-tracked knowledge layer. SHI's Phase-1 role is to place this system behind enterprise controls and prove same-behavior on the deployed platform through registered conformance checks.

This SOW is only for the Phase 1 of the application that has been discussed and aligned. Subsequent phases (2 and 3) will be estimated and priced separately. Further details on these phases are in Appendix E.

2 Project Description

SHI shall provide the following Services to Customer on a fixed fee basis.

2.1 In Scope

Feature ID	Description
F01	Current-state assessment – code/architecture review, SAST/SCA/secrets/container scanning, and DAST. Component and test inventory is confirmed jointly with Customer in Sprint 1; high and critical findings are remediated within Phase 1.
F02	Gap analysis, production-readiness specification, and target architecture.
F03	Containerization of Valtis (engine, advisory app, generators, and the vSAB module), preserving the read-only root filesystem with EFS as the sole writable path, and the isolated, disabled-by-default rerank runtime.
F04	AWS landing zone via MASS – ECR, Fargate, Secrets Manager, EFS (encrypted run/artifact filesystem mounted via access point, enrolled in AWS Backup), RLS Postgres.
F05	Terraform IaC and CI/CD pipeline.
F06	CloudWatch monitoring, logging, and alerting, including ingestion of the engine-emitted per-run token and usage numbers on both production transports, with dashboards and alarms against the DEP5 token budget, and a correction-turnaround metric for the pilot feedback loop.
F07	SSO/SAML authentication (AuthN) with Customer's identity provider.
F08	Authorization (AuthZ) – persona-based RBAC and baseline user management.
F09	Session handling for authenticated multi-user access.
F10	Integrate and verify – Frontdoor: sign-on, program selection, the request-selected-output action defined by the Gate 1 contract, the Convene Advisory Board demonstration surface, and artifact downloads, preserving owner-scoped project memory and change-since-last-session briefing.
F11	Integrate and verify – Honest waiting and failure states (in-progress, empty and failed operations say so explicitly); citations and feedback controls; open display of corrections to knowledge-layer facts with lineage.
F12	Integrate and verify – Owner-scoped run history with artifact download, and the owner-scoped durable session assessment record; regeneration is a new governed run.
F13	Integrate and verify – Output generators: The program deck, to the fixed Gate 1 contract and competitive brief to a fixed contract. The gated output journey calls a Customer-operated generation service over HTTPS (DEP14); Subcontractor integrates and verifies that service on the deployed platform. The Gate 1 output contract pins one committed output (the program deck with references) plus the pre-agreed brief format.
F14	Integrate and verify – Fail-closed check engine: named pass/fail tests; output withheld when a material claim is unsourced, naming the claim and the missing source.
F15	Integrate and verify – PPTX and DOCX exporters; workflow contract, corpus identity and check outcomes recorded in the signed run record accompanying each artifact. PDF export is out of scope for Phase 1.
F16	Integrate and verify – Evidence services, in two parts. (a) Live registry verification: ClinicalTrials.gov and openFDA read-time clients, enabled at deployment behind the DEP6 egress allow-list, verifying trial status and approval facts at generation time. (b) Literature, press/IR and web evidence served from the knowledge layer with dated, source-linked provenance and refreshed through the governed corpus-update process, not fetched per request.

Feature ID	Description
F17	Deploy and demonstrate the Customer-provided simulated advisory board (vSAB) module in full: the sponsor objective becomes a sequenced agenda; the clinician personas suited to each item debate it with deliberately contrarian stances in cross-examination; a non-opining chair closes each item with a consensus-or-split record traceable to counted positions; recommendations carry knowledge-layer citations; and a dissent check confirms the synthesis names each persona who broke from the majority. The demonstration format is recorded in the Gate 1 contract.
F18	Runbook, admin/user guides, architecture document, and project handover.
F19	Migration and conformance verification – execute the registered checks in Exhibit 1 against the deployed system at each milestone; implement and register the six new checks listed in Exhibit 1 Part B against the Customer-provided registry framework, each before the milestone citing it; package machine-readable milestone evidence including raw check outputs.
F20	BIKG integration support – capped, timeboxed integration support (up to ten person-days, Weeks 7–10) for the Customer's parallel BIKG workstream, using Customer-provided access, credentials and evidence-adapter (DEP13). Excluded from all Phase-1 milestones, acceptance criteria and Gate-G2 conditions. Retrieval serving switches to the BIKG-backed path only after it clears the delivered retrieval benchmark with no class regression and licensing for pilot use is confirmed; absent that, it defers to Phase 2 with no schedule or commercial impact. Effort beyond the cap is a change request.

2.2 Deliverables

- Current-state assessment findings, gap analysis and production-readiness specification with target architecture, including the jointly confirmed component and test inventory (F01, F02).
- Secure AWS deployment of the containerized Valtis engine, advisory app, generators and vSAB module via Customer's MASS landing zone, with Terraform IaC, CI/CD pipeline, CloudWatch monitoring and EFS-backed run/artifact storage (F03–F06).
- CloudWatch dashboards and alarms covering run health, per-run token/usage against the DEP5 budget, and correction turnaround (F06).
- SSO/SAML authentication and persona-based authorization with session handling (F07–F09).
- Integrated and verified application layer on the deployed platform: frontdoor, generators via the Customer-operated generation service, fail-closed check engine, PPTX/DOCX exporters with signed run records, and live registry verification (ClinicalTrials.gov, openFDA) (F10–F16).
- The Customer-provided vSAB module deployed and demonstrated in full in the format recorded in the Gate 1 contract (F17).
- Registered conformance checks: the six Part-B checks implemented and registered; machine-readable evidence packages per milestone including raw check outputs (F19).
- BIKG integration support record: hours consumed against the cap and outcome of the benchmark gate (F20).
- Runbook, admin/user guides, architecture document and handover materials (F18).
- Source code for SHI-authored integration and infrastructure work, and the Terraform IaC, delivered to Customer repositories.

2.3 Project Specific Assumptions

Type	Description
Infrastructure	AWS landing zone provided via Customer's MASS program – ECR, Fargate, Secrets Manager, EFS, networking, and egress configured for Phase 1. The Customer-provided Fargate Terraform module creates only the ECS service, one log group and four alarms; SHI extends it within the landing zone.
Infrastructure	The Postgres knowledge layer is provisioned by F04 by the end of Week 2 and loaded jointly with Customer in Week 2–3 (DEP2b), ahead of generator integration. Row-level security enabled.
Infrastructure	Non-GxP scope; no PHI or patient-level data – public and licensed sources only. Single AWS region for Phase 1.
Data, Model & Content	The Valtis scientific core – knowledge layer, workflows, output generators, prompts, rubrics, reasoning framework and the vSAB module – are existing Customer assets. SHI deploys, integrates and verifies them; SHI does not rebuild them.
Data, Model & Content	The following are Customer representations of the delivered system, confirmed at Sprint-1 inventory: a runnable JWT resource server and frontdoor delivering the full F10–F12 behaviors; generators with actionable withholding and PPTX/DOCX rendering; the complete vSAB module; a Postgres knowledge layer with RLS, provenance and correction lineage; a migration kit rehearsed end-to-end twice on the real corpus with fail-closed rejection of corrupted artifacts; a narrow Fargate Terraform module, hardened production+JWT entrypoint, landing-zone requirements sheet, SSO integration spec and monitoring catalog; a 12-check conformance registry with a checked-in exhibit; a minimum of 140-test standalone subset with the minimum of 6,000-test suite in Customer CI; and a 133-query retrieval benchmark with offline replay harness. Material variance adjusts plan and fee through Change Control in either direction.
Data, Model & Content	The frozen pilot corpus provided under DEP2a/DEP2b is a development-and-verification corpus. It supports build, integration, verification and project-team acceptance testing in Phase 1 and is not exposed to users beyond the project team. Before any SME pilot or employee-facing use (Phase 2), the knowledge layer is re-based on Customer's licensed data sources through the same migration kit and re-baselined under the same-behaviour process in this SOW. Customer adds a corpus-designation field to the release manifest and a registered check refusing pilot-mode operation on a development corpus.
Data, Model & Content	Approved LLM access via Customer's AI-Gateway (model-agnostic; two supported production transports), with token budget confirmed by Week 2. SHI does not tune, train or improve models.
Data, Model & Content	The Valtis engine writes per-run token counts and usage numbers to its logs (Customer-provided under DEP11); SHI surfaces these in CloudWatch under F06. The full cost-governance console remains Phase 2.
Data, Model & Content	Prompts, rubrics and fail-closed gates remain under Customer change authority. Configuration changes from the Customer's Phase-1 vSAB tuning track are made by Customer under DEP7 at no cost to this SOW. Each accepted tuning round re-baselines the corpus; verification after that point uses the current baseline plus the standing invariants.
Data, Model & Content	The optional cross-encoder rerank stage is delivered disabled and inert: it only reorders what the governed lexical path retrieves, fails open to default behavior, runs in a runtime isolated from the engine, and performs no model download or network access unless Customer arms it under egress approval. It remains disabled throughout Phase 1. The vector store is an unbuilt seam with no runtime consumer; embeddings ride the migration as inert data.

Type	Description
Data, Model & Content	Source-classification groundwork in the knowledge layer is Customer-side work; SHI preserves the internal/publicly-verified distinction on deployment.
Access, Security & Sources	Access is provided as scoped, read-only access to an approved snapshot of the Valtis application on Customer systems, through Customer's standard vendor-access process (DEP1). This gates the start of the 12-week clock.
Access, Security & Sources	SSO/SAML IdP endpoints are available by Week 3.
Access, Security & Sources	External-source egress is limited to the ClinicalTrials.gov and openFDA public API endpoints (DEP6). No other external retrieval occurs in Phase 1; all other knowledge enters through the governed intake.
Access, Security & Sources	Baseline security scans and SSO are delivered in Phase 1, with high and critical findings remediated within Phase 1; full CISO hardening and penetration-test remediation occur in Phase 2. No pilot users access the deployed system before the security gate completes.
Organizational & Commercial	This SOW is governed by the executed Master Services Agreement between AstraZeneca Pharmaceuticals LP and SHI International Corp.
Organizational & Commercial	One committed output (the program deck) is confirmed, with a pre-agreed alternate format (the brief) fixed at Gate 1. vSAB recommendation-synthesis outputs are provided to facilitated-session participants but are not acceptance-gate evidence.
Organizational & Commercial	Pilot feedback and correction loop: SME feedback enters through the product's correction surface and is triaged jointly, weekly by default. Behavior diverging from documented, verified behavior is a defect; knowledge-content corrections are adjudicated by Customer and applied through the governed correction lineage (SHI does not author knowledge-layer content); all other requests are Phase-2 backlog. SHI's Phase-1 correction obligation covers defects found during Phase-1 project-team acceptance testing and a 30-calendar-day window after Phase 1 Acceptance. Triage and correction during the Gate-G2 pilot period are Phase 2 or the separately scoped support option (O15).
Organizational & Commercial	Customer sponsor and SMEs participate in weekly reviews with feedback within 48 hours of demos. Phase 1 closes on project-team acceptance testing; the formal SME pilot is Phase 2.
Organizational & Commercial	Kickoff requires Fifteen (15) business days' lead time. Signature by September 18 2025 supports a 12th October 2026 kickoff and Phase 1 Acceptance will be 12 weeks post that assuming no other issues/blockers arise.
Organizational & Commercial	Scope is fixed and gated; changes are handled via the Change Control Process. Price covers services only.

2.4 Project Specific Customer Responsibilities

Dependency ID	Description	Owner
DEP1	Provide scoped, read-only access to an approved snapshot of the Valtis application on Customer systems, and access to Customer's AWS landing zone via MASS, on kickoff day, through Customer's standard vendor-access process. Landing-zone ticket includes enrolment of the runs EFS filesystem in AWS Backup with a daily backup plan. Blocks all work if delayed.	Customer IT / MASS
DEP2a	Confirm the frozen pilot corpus (development-and-verification designation) by Week 2 (Gate 1). Congress-source licensing confirmation applies to development-and-verification use in Phase 1; production use from Phase 2 onward rides on Customer's own licensed sources. Blocks the Gate 1 contract if delayed.	Customer Data
DEP2b	Provide the migration kit (export/load/freeze/parity kit and frozen-corpus manifest) and participate in the joint knowledge-layer load in Week 2–3, ahead of generator integration. Blocks generator verification if delayed.	Customer Data
DEP3	Confirm the executed AstraZeneca–SHI Master Services Agreement prior to SOW review. Blocks start if delayed.	SHI + Customer
DEP4	Provide SSO/SAML IdP endpoints and configuration by Week 3. Blocks AuthN/AuthZ if delayed.	Customer IdP team
DEP5	Provide an approved LLM via Customer's AI-Gateway with token budget by Week 2. Blocks generation if delayed.	Customer AI Platform
DEP6	Allow-list egress to the ClinicalTrials.gov and openFDA API endpoints by Week 4. Blocks live registry verification if delayed.	Customer Security / Cloud
DEP7	Maintain content authority over prompts, rubrics, and fail-closed gates on an ongoing basis, and over vSAB configuration (personas, debate behavior, output shape). Blocks output-quality sign-off if delayed.	Customer (Nelson)
DEP8	Ensure Customer sponsor and SME availability for weekly reviews, with feedback within 48 hours of demos. Risks schedule slip if delayed.	Customer (Ross/Nelson)
DEP9	Provide the conformance registry (12 registered checks with checked-in exhibit), the registry framework, handler interface and check-authoring guide, and the definitions of the six Part-B checks (what each proves, handler shape), fixed at Gate 1. Blocks registered-check evidence if delayed.	Customer
DEP10	Provide the vSAB module and its demonstration surface, and agree the demonstration format recorded in the Gate 1 contract. Blocks F17 if delayed.	Customer (Nelson)
DEP11	Provide engine usage reporting – per-run token counts and usage numbers written to logs on both production transports, in time for the F06 dashboard build and the monitoring-and-usage check.	Customer
DEP12	Own and run the Phase-1 vSAB configuration-tuning track and the acceptance of tuning rounds, including re-baselining the frozen corpus on each accepted round.	Customer (Nelson)
DEP13	Provide BIKG access, credentials and the evidence-adapter for the parallel BIKG workstream (F20) by Week 7. Absence defers the workstream to Phase 2 with no schedule or commercial impact.	Customer

Dependency ID	Description	Owner
DEP14	Stand up the generation service interface (HTTPS) and demonstrate readiness at Gate 1, in time for F13 integration. Blocks F13 integration and Milestone 4 if delayed.	Customer

2.5 Out of Scope

Any Services not explicitly listed above as “In Scope” shall be considered out of scope for this project. For example, the areas that are out of scope for this project include, but are not limited to, the following list. If any of these items are required for your organization, they can be scoped separately.

ID	Description	Phase Planned for
001	Source→user entitlement propagation and classification-aware retrieval	Phase 2
002	Conversation audit trail and advanced session/memory governance	Phase 2
003	Token/cost tracking and model-gateway cost governance	Phase 2
004	Guardrails and eval/red-team harness beyond fail-closed checks	Phase 2
005	Governance-aware admin console and usage dashboards	Phase 2
006	SharePoint and BIKG connectors as the production data path. Gate G2 (governed pilot to ≤50 SMEs) is conditioned on the production corpus swap having completed. Capped Phase-1 BIKG support is F20.	Phase 2
007	vSAB governed and multi-user (entitlement, session capture, facilitation) – governance and scale only; the module's scientific behavior is delivered in Phase 1 under F17	Phase 2
008	CISO full hardening and penetration-test remediation (baseline scans occur in Phase 1)	Phase 2
009	Expanded output templates and advanced exporters, including PDF export	Phase 3
010	Semantic cache, analytics, additional source connectors, and pluggable model gateway	Phase 3
011	SAB advanced – multi-agent synthesis and consensus (QB-Agent)	Phase 3
012	Source-data licensing clearance (congress transcripts)	Customer-led
013	Ongoing maintenance, support, and SLA after handover	Separate SOW
014	Rebuilding or re-engineering the Valtis scientific core – knowledge layer, workflows, generators, prompts, rubrics, reasoning framework, or vSAB decision logic	Customer-owned
015	Operating the feedback-tuning loop and pilot-period correction triage – facilitating sessions with pilot users, capturing and triaging structured feedback beyond Phase 1 Acceptance plus 30 days	To be scoped separately
016	Formal SME pilot onboarding and training	Phase 2
017	Vector-store build or dense-retrieval integration; arming the rerank stage or provisioning its model artifacts	Not planned

2.6 Project Duration

Project duration is defined as the entire time taken to complete the project, based on the resources allocated. The estimated project duration is (Phase 1) is **12 WEEKS from kickoff (fixed scope)***.

SHI and the Customer will provide the required resources to deliver this project within the estimated duration. SHI and the Customer will allow for reasonable accommodations due to holidays, vacations, and unforeseen delays in deliveries.

Indicative weekly plan: Discovery, joint inventory confirmation and landing-zone foundation, Weeks 1–2; with the Postgres knowledge layer provisioned by end of Week 2; joint knowledge-layer load, Week 2–3; secure deployment and SSO, Weeks 3–5; application-layer integration and verification (F10–F16), Weeks 4–8; registered-check implementation (F19 Part B), Weeks 3–11; vSAB deployment and demonstration (F17), Weeks 8–11; BIKG capped support (F20), Weeks 7–10; security gate with high/critical remediation, Weeks 9–11; Phase 1 Acceptance, Week 12.

Indicative Sprint Plan: The weekly plan above is broken into six two-week sprints, mapped to the In Scope items (Section 2.1) and Customer Responsibilities (Section 2.4) that gate each sprint, in Appendix F.

** Please be advised that the above timeframe is to provide a general timeline for delivery and is not a true reflection of the total man hours/effort involved for this engagement.*

2.7 Project Management

SHI will provide a Project Manager to work with the Customer to see the engagement through to completion. The SHI Project Manager will cover items such as, but not limited to:

- Manage project scope, schedule and budget as described in this Statement of Work.
- Responsibilities include but are not limited to resource management, status updates, risk management and mitigation strategies.
- Develop and execute change requests as necessary to account for changes in project scope, schedule, and or cost as needed.

2.8 Resources and Skills

SHI will provide individual resources outlined below to be participants for this effort. These resources will participate in all required steps and will be fully or partially responsible for tasks where appropriate:

Title	Role Description	Involvement
Project Manager (Offshore, Weeks 1–12)	Delivery oversight, governance reviews, RAID log, and handover (F18).	Part-time
Solution Architect / Tech Lead (Onshore, Weeks 1–12)	Target architecture, generator and SAB design, Customer/SHI liaison (F02).	Part-time
AI / MLOps Engineer (Onshore, Weeks 1–10)	Assessment, RAG/grounding verification, integration of generators and evidence service, vSAB deployment (F01, F13–F17, F19).	Full-time

Title	Role Description	Involvement
Cloud Integration Engineer (Offshore, Weeks 1–12)	SSO/SAML, RBAC, sessions, service integration (F07–F09).	Full-time
DevOps Engineer (Offshore, Weeks 1–12)	Containerization, MASS landing zone, IaC/CI-CD, CloudWatch including token/usage metrics and alarms (F03–F06).	Part-time
Application Integration Engineer (Offshore, Weeks 3–12)	Integrate and verify the existing application layer on the deployed platform – frontdoor, generators, check engine, exporters, evidence service (F10–F16). BIKG capped support (F20).	Full-time
Front-End Developer (React) (Offshore, Weeks 1–7)	Frontdoor integration and deployment verification, preserving existing behaviors (F10–F12).	Part-time
Data Scientist (Offshore, Weeks 1–5)	Evidence-tiering verification and fail-closed invariant checks (F14, F16).	Part-time
Security Engineer (Offshore, Weeks 5–12)	SAST/DAST, high/critical finding remediation, security gate, SSO/AuthZ hardening support (F01, F07, F08).	Full-time
QA / Verification Engineer (Offshore, Weeks 3–12)	Conformance checks, same-behavior runs, milestone evidence packaging (F19), fail-closed and citation-grounding invariant checks and testing across F03–F17.	Full-time

Peak concurrency is approximately 6 FTE, using a blended onshore/offshore mix; delivery is accelerated by internal engineering automation. The fixed fee covers services only.

2.9 Success Criteria

Each milestone is evidenced by registered conformance checks recorded in Exhibit 1. Milestone evidence cites registered check IDs only; evidence packages include machine-readable raw check outputs, and Customer may re-execute any registered check against the deployed system at each milestone.

1. SOW Signing:

a. Customer and SHI sign this SOW.

2. Gate 1 (~Week 2):

a. Frozen pilot corpus and congress-source licensing confirmed (DEP2a); the consolidated conformance-check list fixed (DEP9); and the vSAB demonstration format recorded in the Gate 1 contract (DEP10); generation-service readiness demonstrated (DEP14).

3. Secure Deployment & SSO Live (~ Week 5):

a. Evidenced by private-origin-verification, deployment-package-contract, startup-and-readiness-gates, image-startup-smoke, image-positive-readiness and the newly registered sso-round-trip check.

4. Integrated Generation & Evidence Verified (~ Week 9):

a. Evidenced by release-manifest-evidence, approved-generator-same-behavior, attribution-registry-readiness, installed-wheel-runtime-smoke, cross-user-access-denial, application-log-content-protection, and the newly registered export-integrity and citation-grounding checks.

5. Phase 1 Acceptance (Week 12):

a. Customer agrees there are no outstanding action items or tasks.

b. Customer agrees all deliverables have been received.

c. Customer signs the Project Acceptance form indicating all project objectives were accomplished.

d. Evidenced by deployed-selected-output-exercise and the newly registered monitoring-and-usage, security-gate and vsab-demonstration checks.

e. Complete evidence pack delivered.

f. Project-team acceptance testing is complete.

g. Scope of the same-behavior run: The same-behavior run is migration evidence: it demonstrates that the deployed system reproduces the system Customer validated, not that outputs never change. As tuning rounds are accepted from Customer's Phase-1 configuration track, the corpus re-baselines to the newly accepted behavior, so improvements land as approved configuration changes rather than acceptance failures. Verification after a re-baseline uses the current baseline plus the standing invariants.

h. Excluded from the acceptance gate: vSAB recommendation-synthesis outputs are provided to participants of every facilitated session in Phase 1, but they are not milestone evidence; the committed acceptance output remains the program deck, with the brief as the agreed alternate format fixed at Gate 1.

3 Assumptions

The project scope and associated price quoted within this Statement of Work are based on the following assumptions. Should any element(s) of these assumptions be lacking during execution of services, additional time and associated fees and expenses may be required to complete this Statement of Work.

1. 1. Minimum lead time for scheduling project kickoff meeting is fifteen (15) business days from our receipt of the signed SOW or fifteen (15) business days from the confirmed start date between SHI and Customer; whichever date is later. Should you require more aggressive scheduling, please contact SHI to determine availability.
2. 2. Knowledge transfer runs throughout the project and in both directions, and is not capped at a fixed number of hours. A joint working session will be scheduled covering knowledge transfer, the Phase-1 feedback-tuning loop with pilot users, the support option, and the rate card. Customer is responsible for providing a resource or resources focused on this project, and the extent of knowledge transfer remains dependent on the availability of those resources.
3. 3. SHI is not responsible for delays caused by failures; including but not limited to systems, personnel or environmental causes or in receiving data from Customer.
4. 4. Any restrictions or requirements regarding the SHI consultants' use of personal equipment must be stated in advance of the commencement of the project.
5. 5. All hardware and/or software and licensing required to perform the above services will be provided by and is the responsibility of Customer. All wiring, hardware, and software required to perform the above services are in working order.

6 6. All parties agree that personnel shall not be asked to perform, nor volunteer to perform, engineering and/or consulting tasks that lie outside the skill sets and experience of personnel. Personnel have the right to decline on a service request if the request falls outside the scope of their experience and expertise.

7 7. Project activity will be scheduled during the hours of 9:00 AM to 5:00 PM local time and also in IST hours for the offshore team. Any work performed outside these hours must be previously agreed upon by both parties and scheduled in advance.

8 8. All Deliverables will be delivered within fifteen (15) business days after the completion of the in-scope tasks or phases of the project. A standard document template will be utilized for this service delivery.

4 Customer Responsibilities

Both Customer and SHI are responsible for the successful execution of this engagement. Prior to the start of this SOW, Customer will indicate to SHI in writing a person to be the point of contact. All project communications will be addressed to such point of contact (the “Customer Contact”). The Customer Contact is responsible for the following:

1 1. Performing a full working backup prior to the commencement of services as SHI is not responsible for lost data.

2 2. Ensuring all related information and communication regarding this project is done through the Project Manager as expeditiously as possible.

3 3. Acting for the Customer in all aspects of the project.

4 4. Making the necessary administrative usernames and passwords available to the designated SHI resource, if required for the successful completion of project.

5 5. Providing detailed and accurate information regarding their current network environment if required for the successful completion of project. This information will include the technical configuration of the domain environment.

6 6. Providing the necessary workspace and network access to provide the above services.

7 7. Providing access to building(s) and room(s) if required for the successful completion of project.

8 8. Obtaining and provide project requirements, information, data, decisions and approvals within one working day of the request, unless both parties agree to a different response time.

9 9. Ensuring that SHI project personnel have reasonable and safe access to the project site and adequate office space, if required.

10 10. Providing technical points-of-contact, who have a working knowledge of the enterprise components to be considered during this project (“Technical Contacts”). SHI may request that meetings be scheduled with Technical Contacts.

11 11. Informing SHI of all access issues and security measures and providing access to all necessary hardware and facilities as required.

12 12. Having the authority to resolve conflicting requirements.

13 13. Helping resolve project issues and ensuring that issues are brought to the attention of the appropriate persons within SHI, if required.

Customer will provide individual resources outlined below to be participants for this project effort. These resources will participate in all required steps and will be fully or partially responsible for tasks where appropriate:

Title	Role Description	Involvement
Sponsor / Project Manager	Project and resource coordination to support the effort as well as authority to make decisions and acceptance at project completion.	Part-time
Technical Contact(s)	Provide access to workspace, building access, and general IT requests related to the effort. May also have responsibility for network, data center and project team activities.	Part-time
Content Authority	Own prompts, rubrics, fail-closed gates and vSAB configuration (DEP7, DEP12); accept tuning rounds and re-baseline the frozen corpus.	Part-time
Data Migration Contact	Provide the migration kit and participate in the joint knowledge-layer load in Week 2-3 (DEP2b).	Part-time

5 Duties of SHI

SHI shall provide the Services and the Deliverables during the term of this engagement in accordance with this SOW and these terms and conditions.

1. SHI will provide all resources, facilities, management, labor, expertise, skills, tools, and equipment necessary for the performance of its obligations under this SOW.
2. Without limiting the foregoing, SHI shall:
   a. keep the Customer Project Manager advised of the progress of the project and the status of the Deliverables;
   b. permit any designated representative of Customer periodically to review the work of SHI personnel performing Services and preparing Deliverables;
   c. perform the Services in a timely manner and provide the Deliverables in accordance with this Statement of Work; and
   d. keep accurate records of work performed on this Statement of Work, evidence of which SHI shall provide to Customer upon request.

6 Change Control Process

The “Change Control Process” is that process which shall govern changes to the scope, schedule or price of the Project during the life of the Project. The Change Control Process will apply to new components and to enhancements of existing components. The Change Control Process will commence at the start of the Project and will continue throughout the Project’s duration.

Under the Change Control Process, a written “Change Request” will be the vehicle for approving any desired changes to the project. It will describe the proposed change, the reason for the change and the effect the change may have on the Project. The requesting party will submit a written Change Request to the other party.

SHI and Customer will review the Change Request. All parties must sign the approval portion of the Change Request to authorize the desired changes. Once signed by all parties, the Change Request will act as an amendment to this SOW.

Sprint 1 inventory confirmation: The plan and fee in this SOW assume the component and test inventory described in Section 2.3. If Sprint 1 discovery materially changes that inventory, the plan and fee are adjusted through this Change Control Process in either direction.

7 Project Initiation Process

Upon receipt of a signed SOW, planning for the project will commence. A key step in the planning process is the kickoff meeting with the Customer.

In the kickoff meeting, the contents of the SOW will be reviewed. This is an opportunity for Customer's team who will be involved with the project to understand the project's goals, tasks, deliverables, and timelines.

Upon completion of the project kickoff meeting, minutes of the kickoff meeting will be created based on the meeting discussion and distributed to Customer. Any changes to the project will be documented in these minutes. If Change Requests are necessary due to these changes, that process will be initiated after the kickoff meeting.

14 Appendices – Supporting Diagrams & Exhibits

14.1 Appendix A: Production Component Architecture

Valtis runs entirely inside Customer's AWS tenant. Storage: EFS run/artifact filesystem (encrypted, mounted via access point). The vector store is an unbuilt seam with no runtime consumer in any planned phase.

[Figure A – Production Component Architecture]

14.2 Appendix B: The Frontdoor & Grounded Output Generation

Frontdoor interface provides program selection, request-selected-output action, and Convene Advisory Board demonstration surface. Generates PPTX and DOCX formats with full evidence trail. PDF export is out of scope for Phase 1.

[Figure B1 – Frontdoor: Program Selection and Convene Advisory Board Actions]

[Figure B2 – Result: Preview, PPTX/DOCX Download, Evidence Trail]

14.3 Appendix C: Output Generation – How a Deck Is Built

Read the evidence lane as: knowledge layer plus live ClinicalTrials.gov and openFDA registry verification. PubMed, press/IR and web evidence enter through the governed corpus path, not per request.

[Figure C1 – Output Generation: Data and Control Flow]

14.4 Appendix D: Simulated Advisory Board (vSAB) – Deployment and Phasing

AZ supplies the full module. Phase 1 deploys, integrates and verifies it; Phases 2–3 add governance and scale – not the science.

[Figure D1 – vSAB Flow and Governance/Scale Additions by Phase]

14.5 Appendix E: Phase 2 & Phase 3 Roadmap (Indicative)

Phase 1 is fixed-scope and fixed-fee, as defined in this SOW. Phases 2 and 3 are shown here for context only – they are indicative, and detailed scope and fixed fee for each will be confirmed at its own gate, following successful delivery of the prior phase.

Phase	Focus	Duration	SHI Fee	Notes
Phase 2	Governance & Multi-User	~10 weeks	Quoted at Gate G2	Includes the production corpus swap (governed load and re-baseline from Customer's licensed sources) as a Gate-G2 condition; SharePoint and BIKG as the production data path; governed multi-user vSAB; formal SME pilot go-live.
Phase 3	Product Maturity & Scale	~8 weeks	Quoted at Gate G3	Expanded templates incl. PDF export; semantic cache; analytics; pluggable gateway; vSAB advanced orchestration; scale toward 500+.

14.6 Appendix F: Indicative Sprint Plan

This table maps the 12-week timeline in Section 2.6 onto six two-week sprints, referencing the In Scope items (Section 2.1) and Customer Responsibilities (Section 2.4) that gate each sprint.

Sprint	Focus	Scope (F-codes)	Dependencies Due	Sprint Review
Sprint 1 (Wks 1–2)	Discovery, Inventory & Foundation	F01, F02; F03, F04 begin; Postgres provisioned by end of Week 2	DEP1 (day 1); DEP3; Gate 1: DEP2a, DEP5, DEP9, DEP10, DEP14	Architecture & readiness spec walkthrough; Gate 1 contract signed
Sprint 2 (Wks 3–4)	Load, environment build-out	DEP2b joint knowledge-layer load (Wk 2–3); F03 completes; F05, F06; F07 begins; F16(a) registry clients; F19 Part B begins	DEP4, DEP6, DEP11	Landing zone + CI/CD pipeline demo
Sprint 3 (Wks 5–6)	Secure Deployment & SSO Live	F07 completes; F08, F09; F10–F13 integration begins against the generation service	None new	Milestone 3 registered checks incl. sso-round-trip

Sprint	Focus	Scope (F-codes)	Dependencies Due	Sprint Review
Sprint 4 (Wks 7–8)	Generation verified	F13–F16 verify; approved-generator-same-behavior; export-integrity and citation-grounding registered; F20 begins	DEP13	Milestone 4 registered checks
Sprint 5 (Wks 9–10)	vSAB, security gate	F17 deploy; high/critical remediation; security-gate and monitoring-and-usage registered; F20 completes	None new	vSAB dry run; security gate evidence
Sprint 6 (Wks 11–12)	Demonstrate, accept, hand over	F17 demonstration; vsab-demonstration registered; deployed-selected-output-exercise; F18; evidence pack	—	Phase 1 Acceptance

14.7 Appendix G: AWS Deployment Topology

Single-region deployment inside Customer's MASS landing zone. Services run on Fargate in private subnets behind the load balancer; secrets in Secrets Manager; runs and artifacts on an encrypted EFS filesystem mounted via access point (AWS Backup daily plan); observability and token/usage telemetry via CloudWatch; egress limited to the two registry API endpoints.

[Figure G1 – AWS Deployment Topology within MASS Landing Zone]

14.8 Exhibit 1: Conformance Checks & Milestone Evidence Map

Part A lists the Customer's delivered conformance registry (DEP9). Part B lists the six checks SHI implements and registers under F19 against the Customer-provided framework, each before the milestone citing it. Milestone evidence cites registered check IDs only; packages include machine-readable raw outputs; Customer may re-execute any check at each milestone.

Part A – Registered Checks (DEP9)

Registered Check ID	What it Proves	Gates
release-manifest-evidence	The deployed release is byte-identical to the approved build	Milestone 4 (Wk 8)
private-origin-verification	The service is reachable only through the approved private origin	Milestone 3 (Wk 5)
cross-user-access-denial	No user can see another user's runs, artifacts or records	Milestone 4 (Wk 8)
application-log-content-protection	Logs never contain content or user-identifying material	Milestone 4 (Wk 8)
installed-wheel-runtime-smoke	The packaged engine runs as installed, not only from source	Milestone 4 (Wk 8)
approved-generator-same-behavior	Generation on the frozen corpus reproduces the validated behavior	Milestone 4 (Wk 8)
startup-and-readiness-gates	Production refuses to start on wrong or ambiguous configuration	Milestone 3 (Wk 5)

Registered Check ID	What it Proves	Gates
deployment-package-contract	The infrastructure keeps its pinned shape (private, read-only, alarmed)	Milestone 3 (Wk 5)
image-startup-smoke	The container image boots	Milestone 3 (Wk 5)
image-positive-readiness	The image reports ready under real production configuration	Milestone 3 (Wk 5)
deployed-selected-output-exercise	The full user journey succeeds on the live deployment	Milestone 5 (Wk 12)
attribution-registry-readiness	The attribution registries generators depend on are loaded and consistent	Milestone 4 (Wk 8)

Part B – Checks to be Incorporated by SHI (F19)

Check ID (to register)	What it Proves	Depends on	Registered by / Gates
sso-round-trip	A Customer identity authenticates end-to-end via the IdP-facing probe in the SSO integration spec and receives a scoped session	DEP4	Wk 5 / Milestone 3
export-integrity	PPTX and DOCX exports open cleanly; the signed run record's workflow contract, corpus identity and check outcomes verify	F15	Wk 8 / Milestone 4
citation-grounding	Engine test-suite invariants promoted to a deployed check: every material claim in released output traces to a dated source	F14	Wk 8 / Milestone 4
monitoring-and-usage	Dashboards and alarms live and reporting per-run token/usage against the DEP5 budget on both transports	DEP11, F06	Wk 12 / Milestone 5
security-gate	F01 scans complete and high/critical findings remediated; precedes any pilot-user access	F01	Wk 12 / Milestone 5
vsab-demonstration	The Gate-1 demonstration format executes on the deployed module: agenda, cross-examination, chair record traceable to counted positions, dissent check	DEP10, F17	Wk 12 / Milestone 5